Ideal Gas Processes

In most applications, gases are used to transfer energy. For ideal gases, non flow processes are most commonly analysed. We usually analyse the PVT equation of state(using combined gas law), the PV diagram, the change in internal energy, the heat transferred and the work done.

There are 5 standard non flow processes: Isobaric, Isochoric, Isothermal, Adiabatic, Polytrophic.

Isobaric Processes

Isobaric refers to a constant pressure process.

One example of an isobaric process is heating the gas inside a piston cylinder device, while the piston moves upwards as the gas maintains the same pressure.

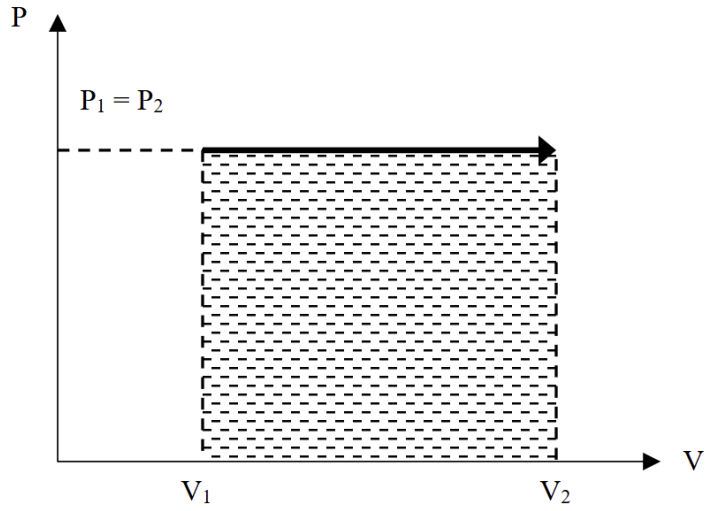
As pressure is the same, the PVT equation of state is

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

This equation is also known as Charles' Law, which states that at a constant pressure, the volume of a fixed mass of gas is proportional to its absolute temperature(temp in kelvin)

$[V_1 = \frac{V_2}{T_2} \times T_1 \Rightarrow V_1 = kT_1 \Rightarrow V_1 \propto T_1]$

On a PV diagram, an isobaric process is represented by a horizontal line. Shown is an expansion isobaric process, as volume increases:



The work done during an isobaric process can be found by:

$$W = P(V_2 - V_1)$$

Notice how this formula also matches with the area under the PV diagram. Hence, we can generalise work as:

$$W = \int_{V_1}^{V_2} P \, dV$$

Notice how the gas expands, hence ΔV is positive and work done will be positive, meaning work is output. If the gas underwent compression, ΔV is negative, and work is input.

We can also rewrite the above formula as so:

$$W = PV_2 - PV_1$$
$$W = mRT_2 - mRT_1$$
$$W = mR(T_2 - T_1)$$

Using this formula and Joules Law of $\Delta U = mC_v(T_2 - T_1)$, which is used for any non flow ideal gas processes, we can find the heat transferred:

$$
\begin{gathered}
Q - W = \Delta U \\
Q - mR\Delta T = mC_v\Delta T \\
Q = mC_v\Delta T + mR\Delta T \\
Q = m(C_v + R)\Delta T \\
Q = m(\cancel{C_v} + C_p - \cancel{C_v})\Delta T \\
\underline{Q = mC_p\Delta T}
\end{gathered}
$$

The heat transferred can also be found with the work done formula of volume:

$$
\begin{gathered}
Q - W = U_2 - U_1 \\
Q = U_2 - U_1 + W \\
Q = U_2 - U_1 + PV_2 - PV_1 \\
Q = U_2 + PV_2 - (U_1 + PV_1) \\
Q = H_2 - H_1 \\
Q = \Delta H
\end{gathered}
$$

Isochoric Processes

Isochoric refers to a constant volume process.

One example is when a gas is heated in a container with solid walls, forcing the volume to stay constant.

As volume is constant, the PVT equation of state is

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

This is known as the pressure law, or the Gay-Lussac's Law, which states for a gas of constant volume and fixed mass, the pressure is directly proportional to the absolute temperature.

On a PV diagram, isochoric processes are represented by vertical lines.

As volume does not change, the work done is 0.

Notice how the area under the graph still represents the work done, as the area is 0.

Hence, we know that $Q = \Delta U$. Using Joules' Law:

$$Q = \Delta U = mC_v(T_2 - T_1)$$

Isothermal Processes

Isothermal refers to a constant temperature process.

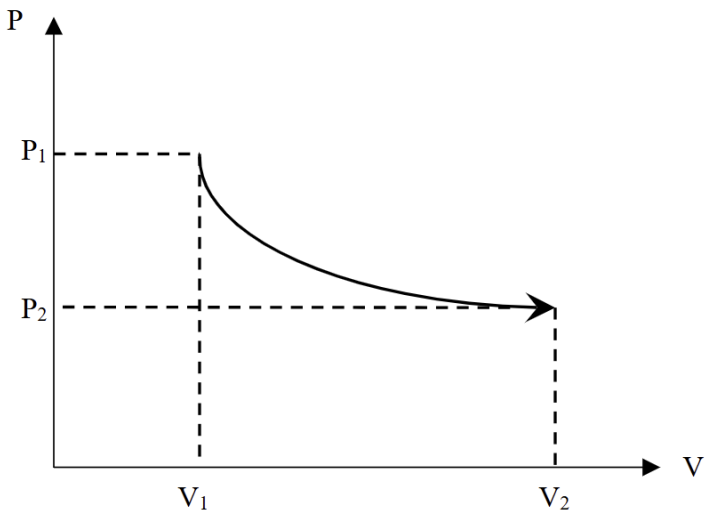
One example is when a gas is compressed while inside a container that allows heat transfer (non insulated walls). The gas is able to lost heat as its compressed to maintain thermal equilibrium with the surroundings, hence making it an isothermal process.

The PVT equation of state is

$$P_1V_1 = P_2V_2$$

This equation is also known as Boyle's Law, which states a fixed mass of gas at a constant temperature has its pressure inversely proportional to its volume.

On a PV diagram, an isothermal process has a curved line representing the process:



From this diagram, we can determine the work done by a isothermal process by integrating respect to volume:

$$PV = mRT \text{ [RHS is constant]}$$
$$P = \frac{mRT}{V}$$
$$\therefore W = \int_{V_1}^{V_2} \frac{mRT}{V}\, dV$$
$$W = mRT \int_{V_1}^{V_2} \frac{1}{V}\, dV$$
$$W = mRT\, [\ln(V)]_{V_1}^{V_2}$$
$$W = mRT(\ln(V_2) - \ln(V_1))$$
$$\underline{W = mRT \ln\left(\frac{V_2}{V_1}\right)}$$

which can be written as:

$$W = P_n V_n \ln\left(\frac{V_2}{V_1}\right)$$

If we use Joules' Law to determine ΔU, we realise that the ΔT term is 0, as the process is isothermal. Hence,

$$\Delta U = mC_v(0)$$
$$\Delta U = 0$$

Thus, heat transferred can be determined as

$$Q - W = 0$$
$$Q = W$$
$$Q = mRT \ln\left(\frac{V_2}{V_1}\right) = P_n V_n \ln\left(\frac{V_2}{V_1}\right)$$

Adiabatic Processes

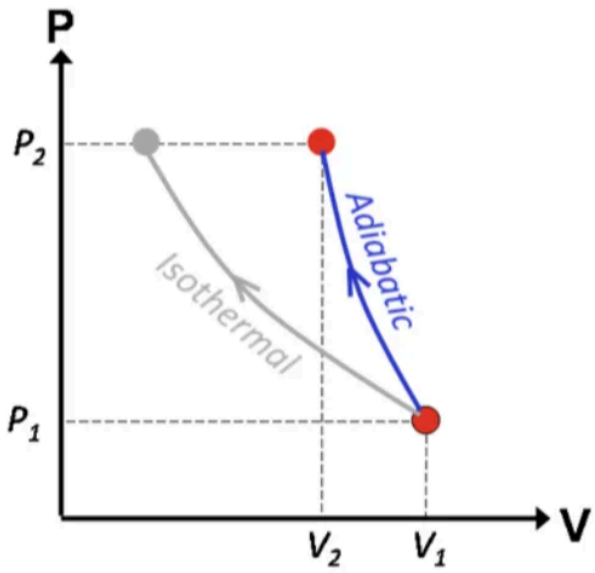
Adiabatic *(ay-dea-batic)* is defined as a process with no heat transfer.

To achieve an adiabatic process, there are 2 means of doing so:

Conduct the process in a well insulated system, or conduct the process extremely rapidly so there is insufficient time for heat transfer to occur.

So for a piston cylinder example, the piston must be moved rapidly to reduce heat transfer, or must have the cylinder walls well insulated.

An adiabatic process on a PV diagram is represented by a curved line, steeper than an adiabatic process.



The PVT equation of state is given as

$$P_1V_1^\gamma = P_2V_2^\gamma$$

where gamma is the specific heat ratio, or $\gamma = \frac{C_p}{C_v}$

Recall that the specific heat ratio is always greater than 1, hence causing the process on the PV diagram to be more curved.

Recall that the combined gas law of $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$ will still apply to adiabatic processes. Hence, the following can be derived:

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

Since we know that $PV^\gamma = \text{constant}$, we can make P the subject in the equation $PV^\gamma = k$

-

$$
\begin{aligned}
P &= \frac{k}{V^\gamma} \\
W &= \int_{V_1}^{V_2} \frac{k}{V^\gamma}\, dV \\
&= k \int_{V_1}^{V_2} V^{-\gamma}\, dV \\
&= k \left[\frac{V^{-\gamma+1}}{-\gamma+1} \right]_{V_1}^{V_2} \\
&= -k \left[\frac{V^{-\gamma+1}}{\gamma-1} \right]_{V_1}^{V_2} \\
&= -k \left(\frac{V_2^{-\gamma+1}}{\gamma-1} - \frac{V_1^{-\gamma+1}}{\gamma-1} \right) \\
&= \frac{kV_1^{-\gamma+1}}{\gamma-1} - \frac{kV_2^{-\gamma+1}}{\gamma-1} \\
&= \frac{P_1 V_1^\gamma V_1^{-\gamma+1}}{\gamma-1} - \frac{P_2 V_2^\gamma V_2^{-\gamma+1}}{\gamma-1} \\
&= \frac{P_1 V_1}{\gamma-1} - \frac{P_2 V_2}{\gamma-1} \\
\underline{W} &\underline{= \frac{P_1 V_1 - P_2 V_2}{\gamma-1}}
\end{aligned}
$$

which can also be rewritten as

$$W = \frac{mR(T_1 - T_2)}{\gamma - 1}$$

And since no heat is transferred during this process,

$$-W = \Delta U$$

Thus the change in internal energy can be found either from Joules' Law or the work done.

Polytrophic Processes

Polytrophic refers to a process with none of the quantities preserved, aka a general process.

An example of a polytrophic process is when a gas cylinder is compressed by a piston at a moderate speed. It is still able to transfer heat, but still

happens fast enough that some heat transfer is minimised, hence being neither an isothermal nor an adiabatic process.

The general PVT equation of state is

$$P_1V_1^n = P_2V_2^n$$

where n is the polytrophic index, which can be negative or infinity.

With different values of n, this formula can be used to represent all different processes.

when n = 0, the process is isobaric

when n = 1, the process is isothermal

when n = γ, the process is adiabatic

when n = ∞, the process is isochoric

From the example earlier, since the piston was compressed at moderate speed, its value of n will be greater than 1 but less than the specific heat ratio.

We can actually use the same adiabatic formulas from earlier to find PVT equations of state for polytrophic processes:

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{n-1} = \left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}}$$

as well as the work done:

$$W = \frac{P_1V_1 - P_2V_2}{n-1} = \frac{mR(T_1 - T_2)}{n-1}$$

and since we can find ΔU with Joules Law, heat transferred can also be found:

$$Q - W = \Delta U$$
$$Q = W + \Delta U$$
$$Q = \frac{P_1V_1 - P_2V_2}{n-1} + mC_v(T_2 - T_1) = \frac{mR(T_1 - T_2)}{n-1} + mC_v(T_2 - T_1)$$